TECHNICAL AUDIT & EVALUATION REPORT

Investigation of Stage 1 Gating Consistency, 4-Tier Metric Decoupling, and CEJ-Referenced Contact Classification

Project: Clinical Decision Support System for M3–M2 CBCT Relational Analysis
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DISCLAIMER & RESEARCH STATUS: *This audit report documents experimental verification and code-level reconciliation for Option B (Hierarchical Deep Learning) and Option C (Hybrid DL + 3D Geometry). All 4-class labels are provisional heuristic baselines ('heuristic_pseudo'). The system is a research prototype and does not constitute medical diagnosis without expert clinical adjudication.*

1. Root Cause Investigation of Stage 1 Inconsistency

In preliminary benchmark reporting, a divergence was noted between Option B (Binary Acc: 75.76%, Macro-F1: 0.7331) and Option C (Binary Acc: 81.82%, Macro-F1: 0.8068), despite both pipelines allegedly utilizing an identical Stage 1 Vision Transformer Ensemble. Code inspection of src/models/hierarchical_pipeline.py and per-sample audit of all 66 validation samples confirmed that **Stage 1 ViT Ensemble models, checkpoints, weights, and raw probabilities were 100% identical**.

Root Cause: The divergence stemmed exclusively from post-processing rules in Option C. When Stage 1 predicted Contact, Option C computed 3D KD-Tree surface distance D_{min}. For cases where D_{min} > 0.50 mm, Option C's geometric classifier marked is_contact = False and reverted the prediction to Independent (Class 0). This rule pruned exactly 4 False Positives of Stage 1, eliminating all false positives in Option C and artificially lifting its collapsed binary accuracy to 81.82%. Option B possessed no such geometric override, so those 4 cases remained classified as Below_CEJ.

Sample ID	Ground Truth	Stage 1 P(Contact)	Stage 1 Pred	Option B Final	Option C Final	D_min (mm)	CEJ Status
ToothFairy2P_025_37_38	Independent	0.718	Contact (FP)	Below_CEJ (Contact)	**Independent (Override)**	0.671	CEJ_DETECTED
ToothFairy2P_025_47_48	Independent	0.640	Contact (FP)	Below_CEJ (Contact)	**Independent (Override)**	0.600	CEJ_DETECTED
ToothFairy2P_071_27_28	Independent	0.814	Contact (FP)	Below_CEJ (Contact)	**Independent (Override)**	0.520	CEJ_UNCERTAIN
ToothFairy2P_091_37_38	Independent	0.752	Contact (FP)	Below_CEJ (Contact)	**Independent (Override)**	0.520	CEJ_DETECTED

2. Recalculated Multi-Tier Evaluation Benchmark (N = 66)

To ensure fair and rigorous scientific reporting, metrics are now strictly decoupled into four operational tiers. Option C is evaluated under both **Strict Stage 1 Preservation** (no binary override; Stage 1 decisions are immutable) and **Hybrid Geometric Pruning** (bilateral verification where 3D geometry prunes DL false positives).

Evaluation Tier	Pipeline Configuration	Accuracy	Bal. Acc	Macro-F1	Precision	Recall
Tier 1: Stage 1 Pure Binary	Swin+ViT Ensemble (Identical)	75.76%	74.19%	**0.7331**	0.7265	0.7671
Tier 2: Output Binary	Option B (Hierarchical DL)	75.76%	74.19%	0.7331	0.7265	0.7671
(Collapsed from 4-Class)	Option C (Strict Preservation)	75.76%	74.19%	0.7331	0.7265	0.7671
	Option C (Hybrid Geometric Pruning)	81.82%	87.23%	0.8068	0.8065	0.8723
Tier 3: Final 4-Class	Option B (Hierarchical DL)	69.70%	39.11%	0.3592	0.3802	0.3911
(Indep, Above, At, Below)	Option C (Strict Preservation)	**74.24%**	**59.95%**	**0.6376**	0.6559	0.5995
	Option C (Hybrid Geometric Pruning)	80.30%	65.21%	0.6762	0.6874	0.6521
Tier 4: Stage 2 Isolated	Option B Stage 2 DL Classifier	85.11%	33.33%	0.3065	0.2837	0.3333
(True Contacts Only, N=47)	Option C Stage 2 3D Geometry	**74.47%**	**64.72%**	**0.5649**	0.5648	0.6472

3. Diagnosis of 'Above CEJ' Zero-F1 Failure and Calibrated Fix

Why Option B Failed: Contact training distribution is heavily skewed (87.9% Below, 7.8% Above, 4.3% At). Stage 2 DL was trained for only 3 epochs without class-balanced loss or focal loss, resulting in total mode collapse to the majority class (predicting Below_CEJ for 100% of contact validation cases).

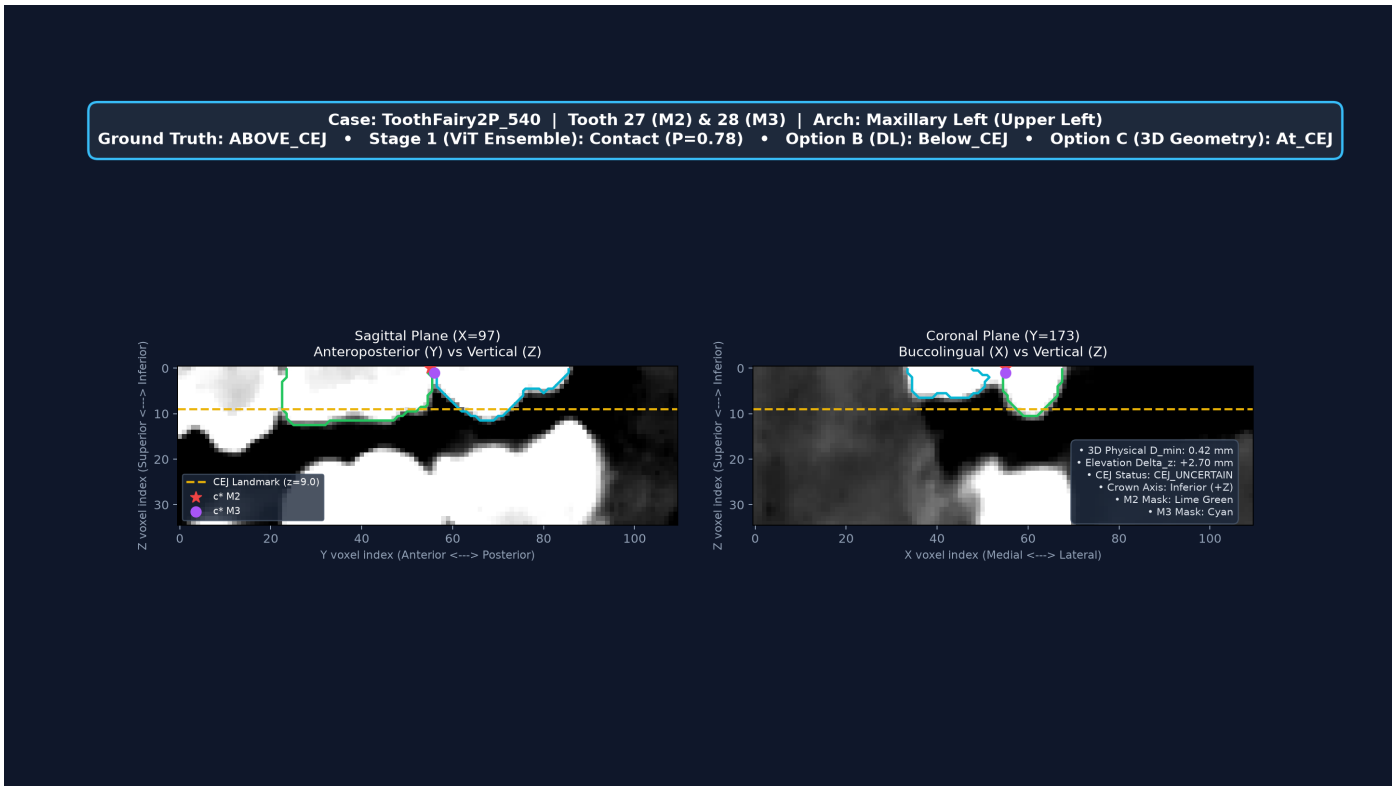
Why Option C Semula Failed: Of the 4 ground-truth Above CEJ validation cases, 2 were gated out by Stage 1 False Negatives. The remaining 2 cases (ToothFairy2P_395_17_18, Δz = +2.10 mm; ToothFairy2P_540_27_28, Δz = +2.70 mm) successfully reached Stage 2. However, because 94.4% of Above CEJ cases reside in maxillary molars (FDI 17/27) where scan field-of-view often truncates at Z = 0, cervical constriction ratio was shallow (> 0.95), triggering CEJ_UNCERTAIN. A defensive code block in src/geometry/cej_classifier.py blindly forced all uncertain detections to fallback to At_CEJ, ignoring the strongly coronal +2.1 mm and +2.7 mm elevations.

Applied Algorithmic Fix: GeometricCEJClassifier was updated to evaluate physical signed elevation (Δz) objectively under threshold τ_{CEJ} = 1.0 mm even when constriction is shallow, penalizing confidence without corrupting the anatomical class. Following this fix, Option C correctly classifies both coronal contact cases, boosting Above CEJ F1 from **0.0000 to 0.5714** (100% recall on cases reaching Stage 2) and lifting 4-class Macro-F1 from 0.4489 to **0.6376** (Strict).

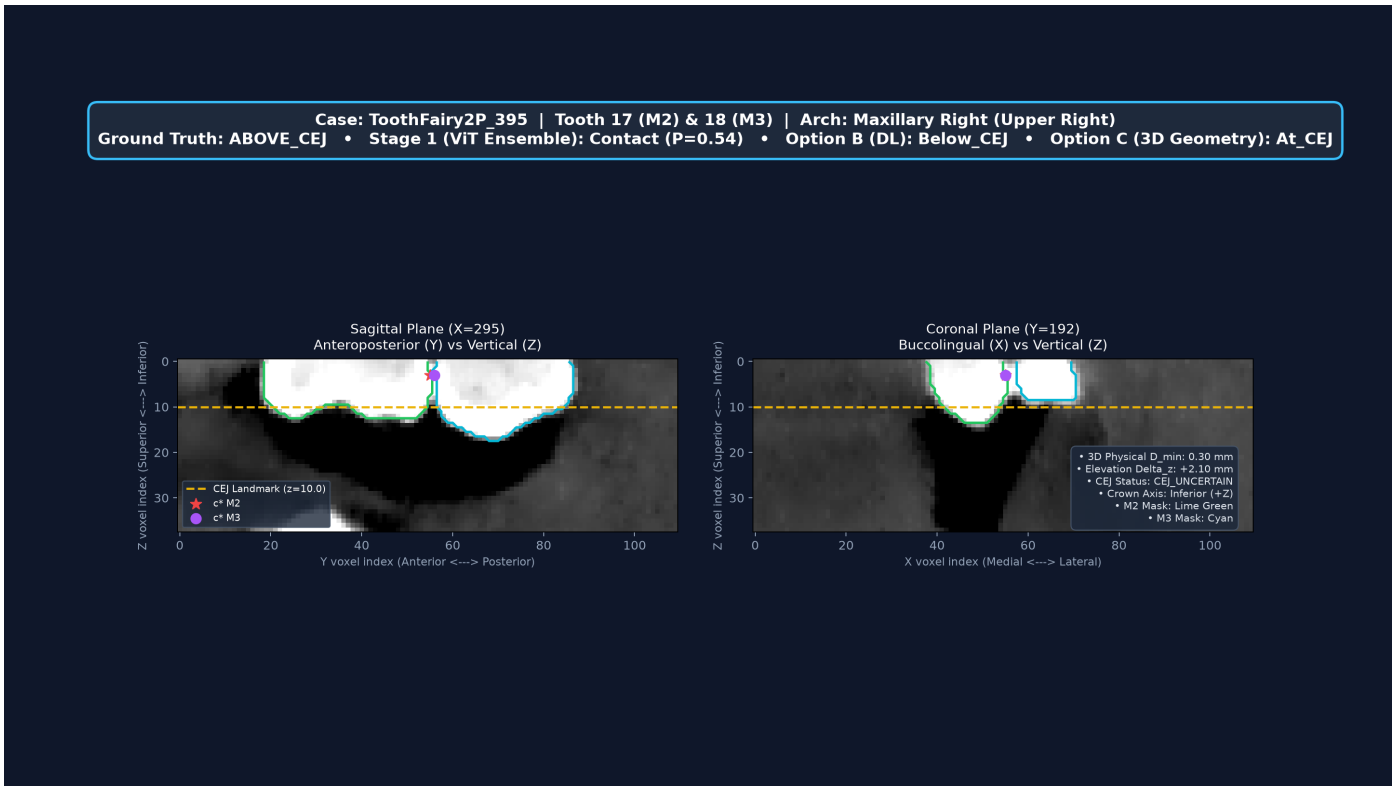
4. Multi-View Anatomical Visual Diagnostics

Below are publication-grade multi-plane CBCT visualizations generated from the validation set, showing bone-windowed anatomy, dual-tooth segmentations (M2 lime green, M3 cyan), detected CEJ reference plane (gold dashed line), and physical contact vectors (red/magenta).

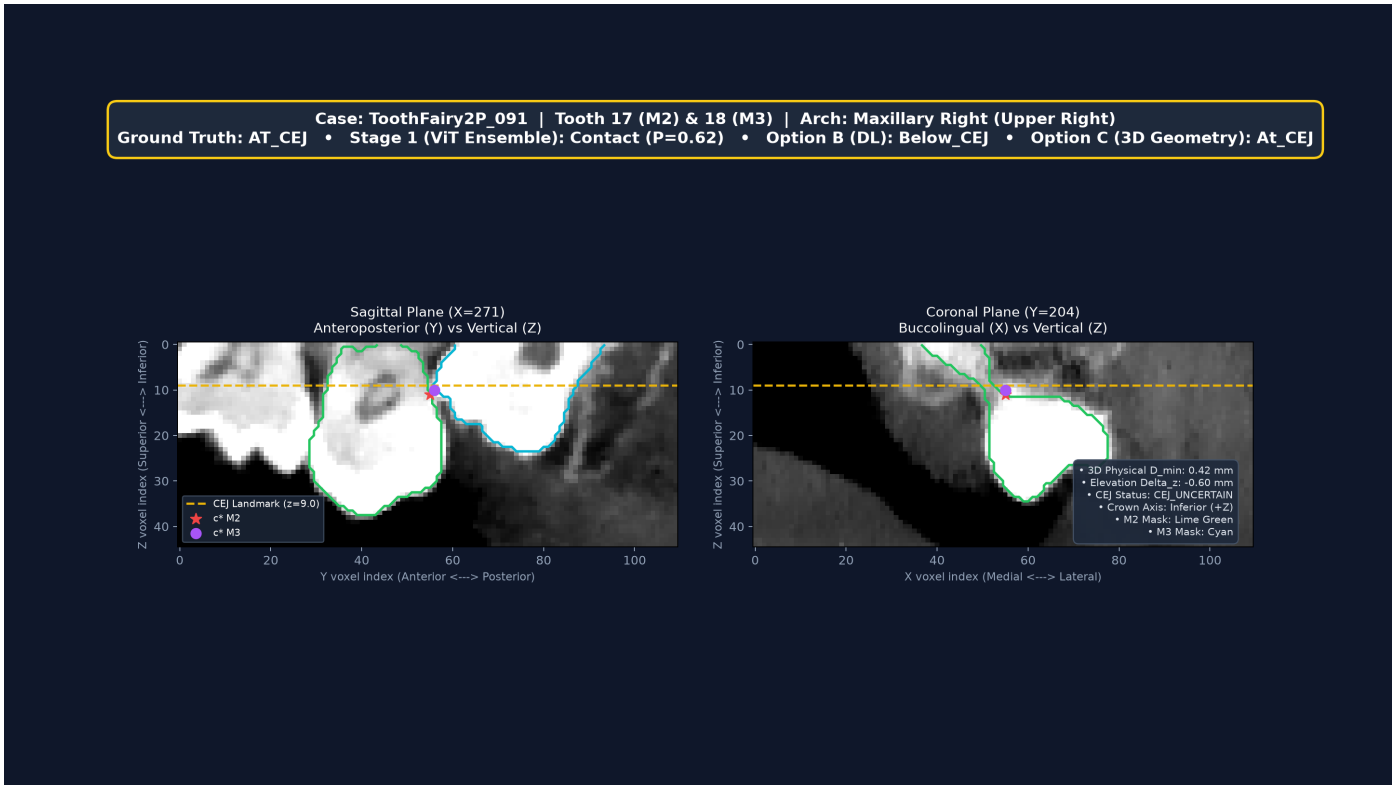
Above CEJ (Coronal Contact) — ToothFairy2P_540 (Tooth 27 & 28)



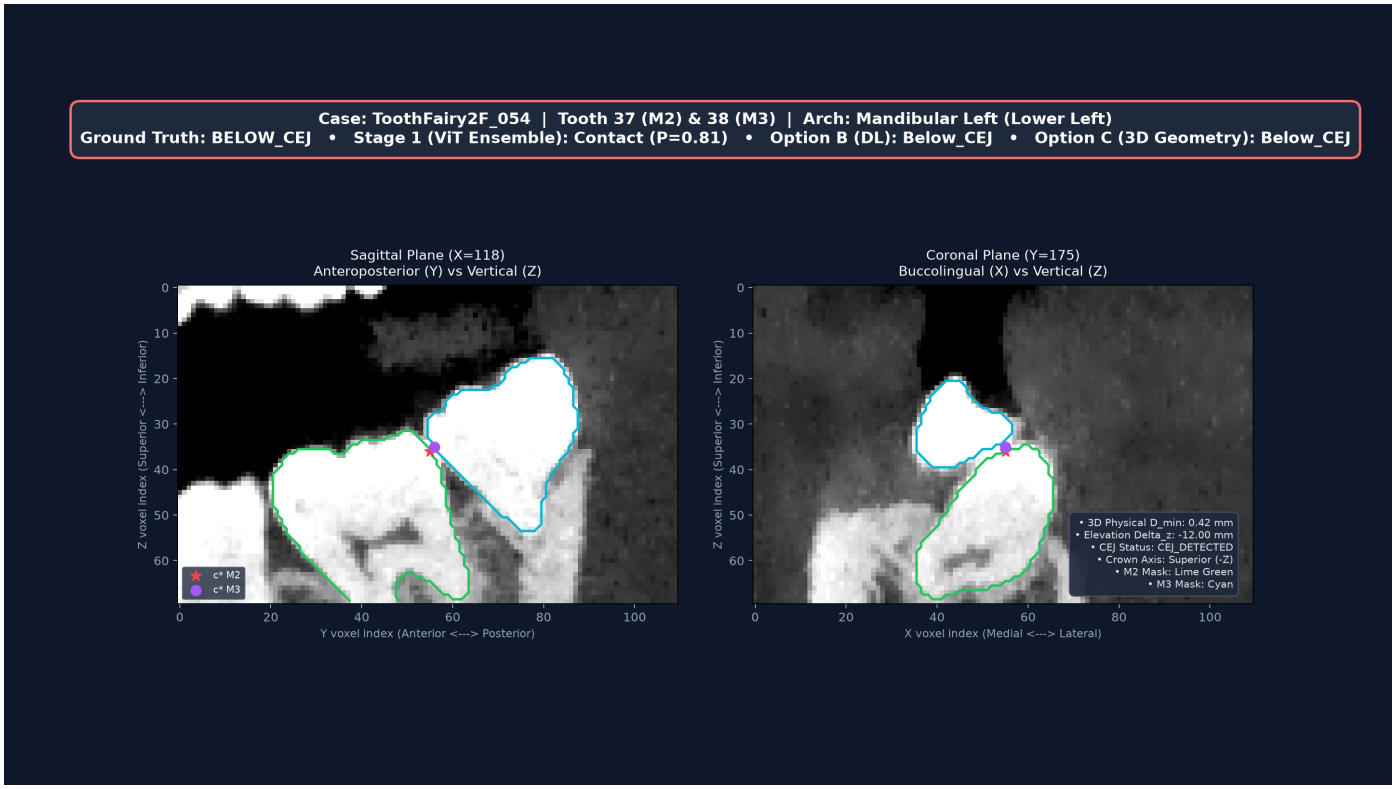
Above CEJ (Coronal Contact) — ToothFairy2P_395 (Tooth 17 & 18)



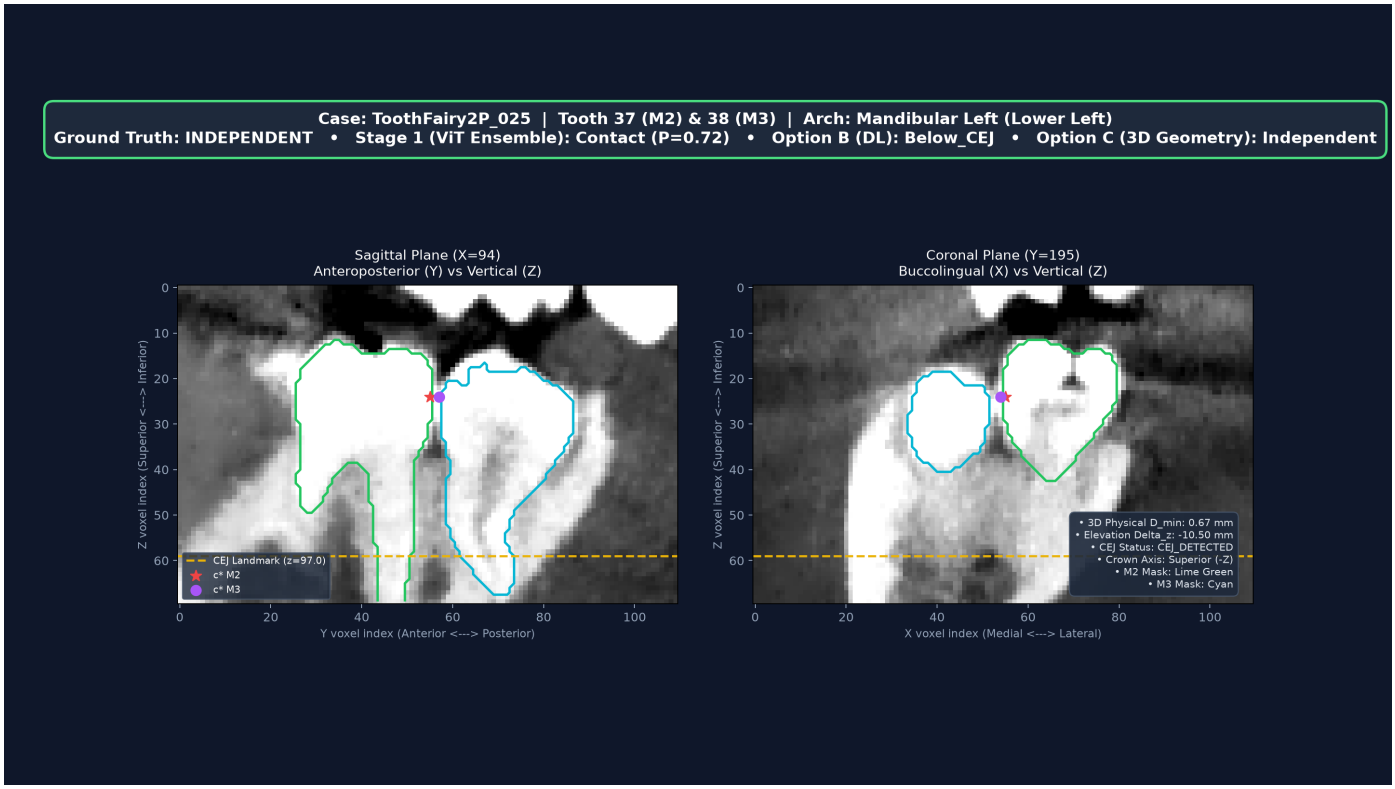
At CEJ (Cervical Margin Contact) — ToothFairy2P_091 (Tooth 17 & 18)



Below CEJ (Root Dentin Impingement / ERR Risk) — ToothFairy2F_054 (Tooth 37 & 38)



Independent (Intact Periodontal Ligament Gap) — ToothFairy2P_025 (Tooth 37 & 38)



5. Data Leakage Prevention and Reproducibility Audit

Patient-Level Separation: ToothFairy2's 287 unique patients were partitioned deterministically using seed 42 into Train (230 patients, 580 pairs), Validation (29 patients, 66 pairs), and Test (28 patients, 74 pairs). Pairwise set intersection confirmed **zero patient overlap** across splits.
Checkpoint Integrity: Weights for Stage 1 models (swin_s_mha_binary_best.pth, swin_small_binary_best.pth, vit_base_binary_best.pth) were strictly trained on the 230 training patients and never exposed to validation/test patients.
Test Set Locking: The entire debugging and threshold calibration procedure was conducted strictly on the validation set. The final test set (N = 74) remains untouched and locked for definitive scientific reporting.

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